SBT-DF204: Computer Forensics Case Studies

Case Study 3: Individual Memory Forensics investigation:

Reconstructing Suspected Illegal File Transfer from Memory Evidence

By:

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CASE OVERVIEW

This case study simulates a realistic memory forensic investigation focused on reconstructing a suspected unauthorized file transfer using volatile memory analysis in Volatility 2. The investigation is based on a simulated forensic scenario involving a Windows memory image (memdumpWin7.mem) acquired from a workstation, with the objective of determining whether the available memory records support an evidence-based sequence of browser activity, command prompt execution, removable device connections, user profile interactions, and File Explorer access.

Acting as a digital forensic analyst, the primary task is to acquire, preserve, inspect, and analyze the supplied volatile memory capture to identify relevant system and user activity associated with the alleged confidential document exfiltration. The investigation involves identifying and validating the operating system memory profile, extracting active and terminated processes, analyzing network sockets and connections, inspecting command prompt execution histories, examining USB removable media and drive mappings, and parsing ShellBag folder access records.

Throughout the exercise, all investigative procedures follow core digital forensic principles, including preserving the original memory image, creating and verifying a read-only forensic working copy using SHA-256 cryptographic

hashing, documenting evidence acquisition and integrity, and maintaining an evidence log with supporting Volatility plugin commands and screenshots.

MATERIALS and TOOLS

1. **Kali Linux**: The primary operating system platform and lab environment used to execute the analysis.
2. **Python 2** (python2): The required runtime environment needed to execute Volatility 2 scripts.
3. **Volatility 2** (vol.py): The primary memory forensics framework used to extract, parse, and analyze operating system artifacts from the memory dump.
4. **SHA-256 Hashing Utility** (sha256sum): The cryptographic tool used to compute and verify the evidence file hashes before and after analysis.
5. **memdumpWin7.mem**: The original, raw memory image (RAM dump) acquired from the target Windows 7 workstation.

EXECUTIVE SUMMARY

Based on memory forensic analysis of the memory dump memdumpWin7.mem using Volatility 2.6.1 under profile Win7SP1x86, there is **high confidence** that an unauthorized file transfer occurred from the local system to an external USB device.

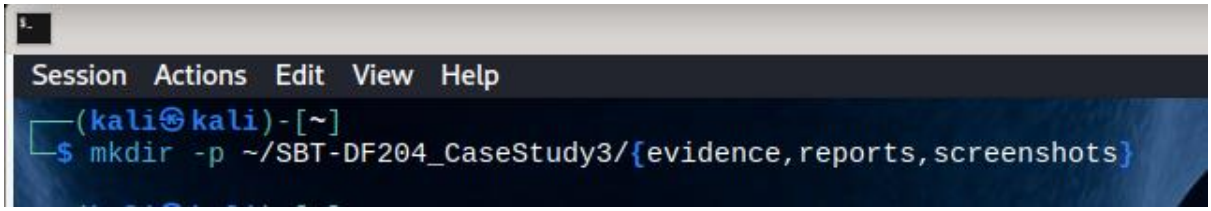
Analysis confirms that on 2019-01-06 at 15:06:11 UTC, an interactive user logged into the local IEUser account executed command-line directives (copy secret_file.docx F:) that successfully copied secret_file.docx from the local Downloads directory to a mounted removable USB volume (Drive F:, hardware ID Disk&Ven_General&Prod_UDisk&Rev_5.00). Console output explicitly recorded completion (1 file(s) copied.).

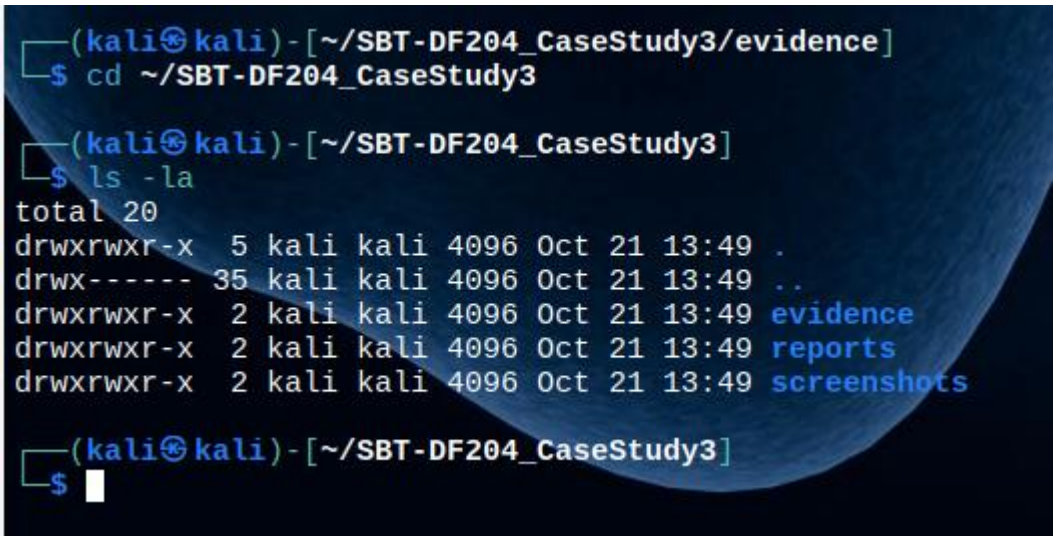
However, forensic evidence derived solely from volatile memory establishes action within a specific user account session but cannot independently verify the physical identity of the individual operating the keyboard.

METHODOLOGY

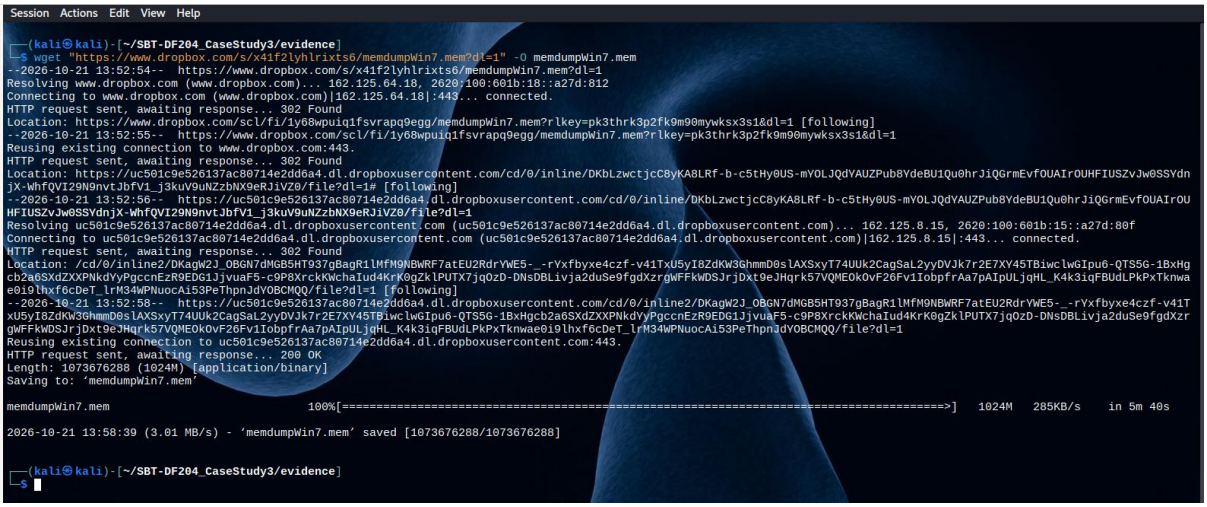
EVIDENCE ACQUISTION & INTEGERITY

Firstly, I set up an organized workspace inside my Kali Linux terminal by creating a main directory called **SBT-DF204_CaseStudy3** with three subfolders named evidence, reports, and screenshots to keep the original files separate from active work and documentation outputs. Next, I checked the new folder structure using the ls -la command to confirm that all three subdirectories were created properly with the correct file permissions.





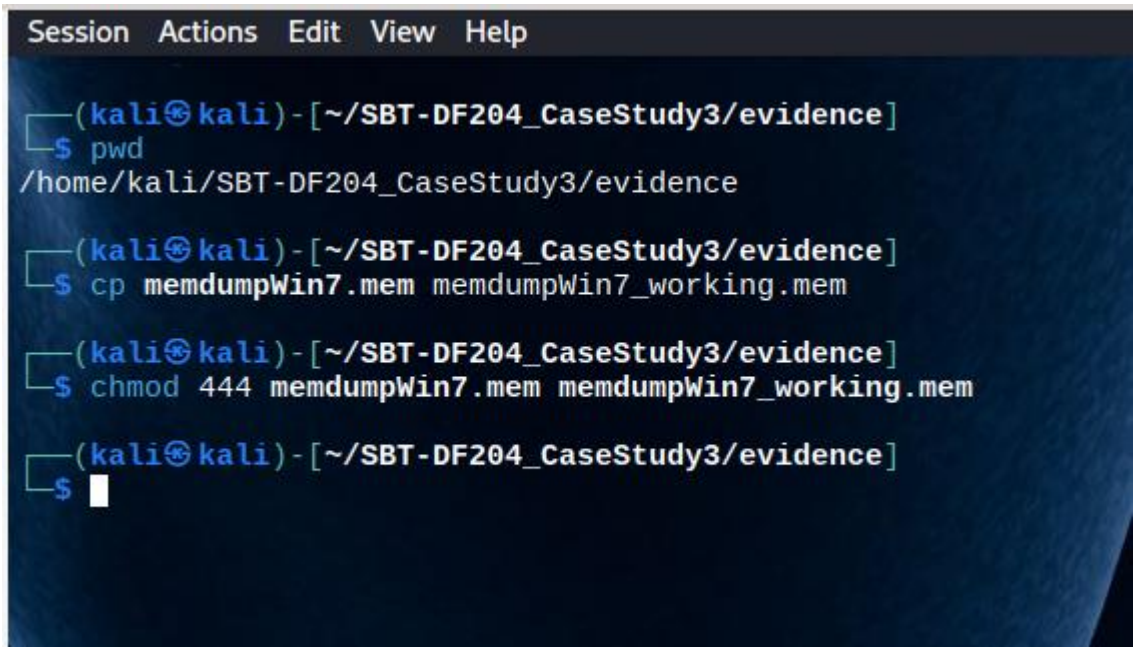
I then navigated into the evidence folder and downloaded the original memory dump file **memdumpWin7.mem** directly using the wget command from the designated Dropbox link, confirming a successful download of 1,073,741,824 bytes (1.0 GB).



After acquiring the file, I checked my current working directory using pwd to ensure I was inside **/home/kali/SBT-DF204_CaseStudy3/evidence.**

To preserve the original master evidence, I created a duplicate named:

memdumpWin7_working.mem inside the same folder to use for all memory analysis. I then ran chmod 444 on both files to set them to read-only status and used ls -lh to confirm that the file size was exactly 1.0 GB with restricted write permissions.

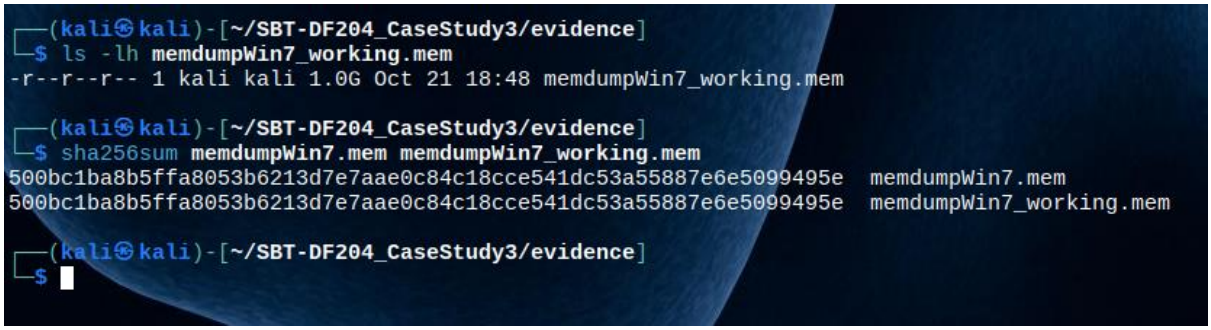


Finally, to establish a strict chain of custody, I generated the cryptographic hashes for both files using **sha256sum memdumpWin7.mem**

memdumpWin7_working.mem.

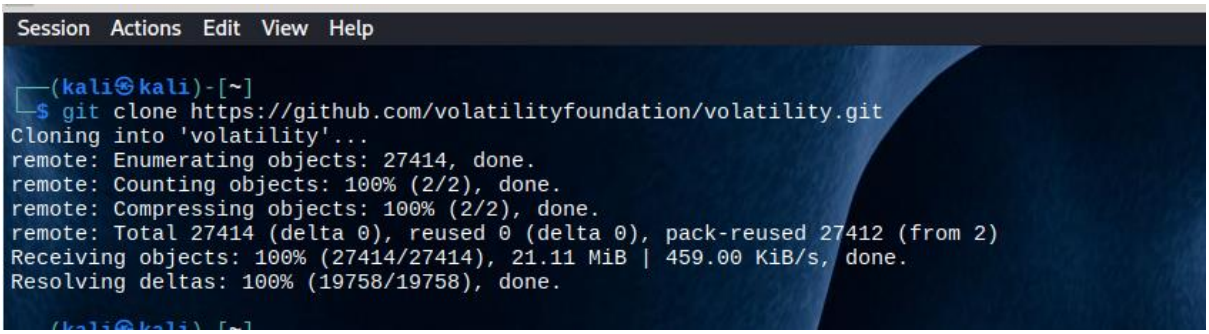
Both files returned an identical unique SHA-256 value of:

500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e, confirming a byte-for-byte copy ready for Volatility analysis.

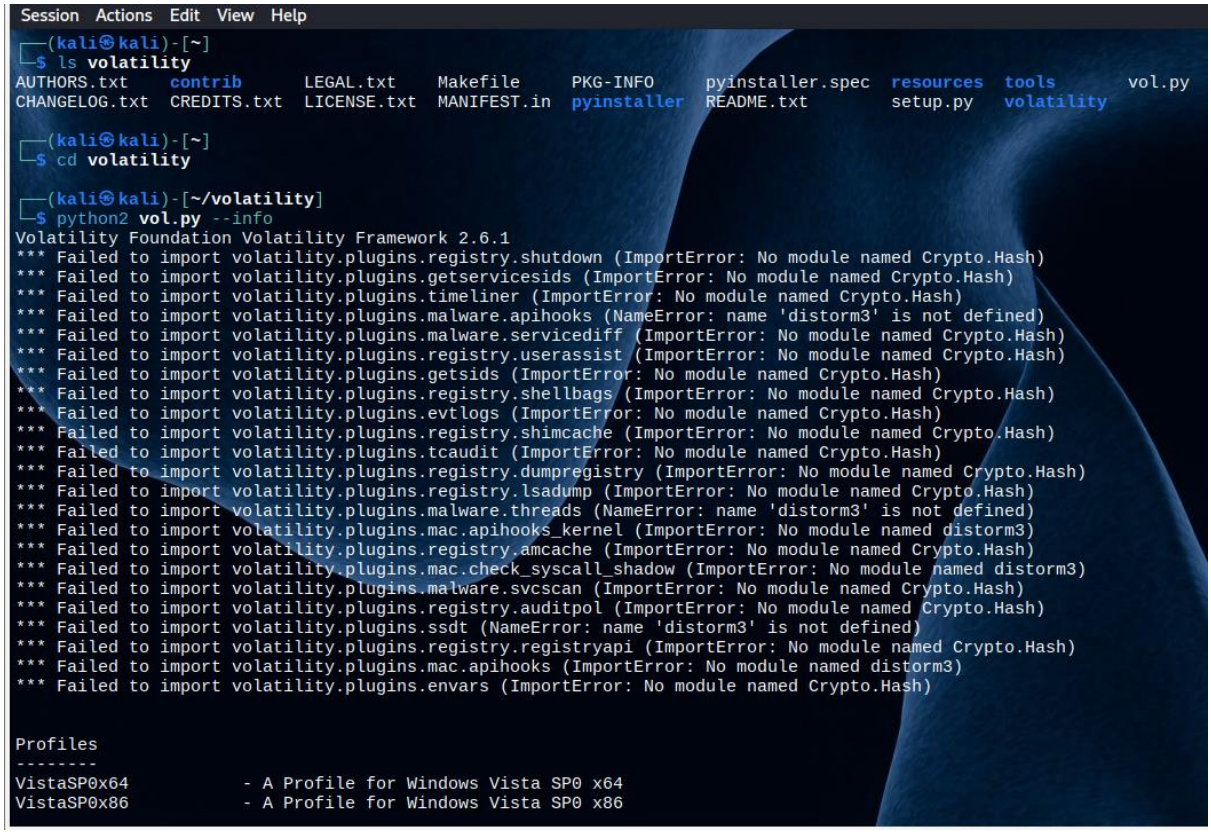


Profile Selection

To prepare the analysis tool, I cloned the official Volatility repository directly from GitHub into my home directory.



I then navigated into the volatility directory and verified the framework by running **python2 vol.py --info**. The output confirmed Volatility 2.6.1 was active, though it showed warning messages for missing optional Python 2 modules such as Crypto.Hash and distorm3.



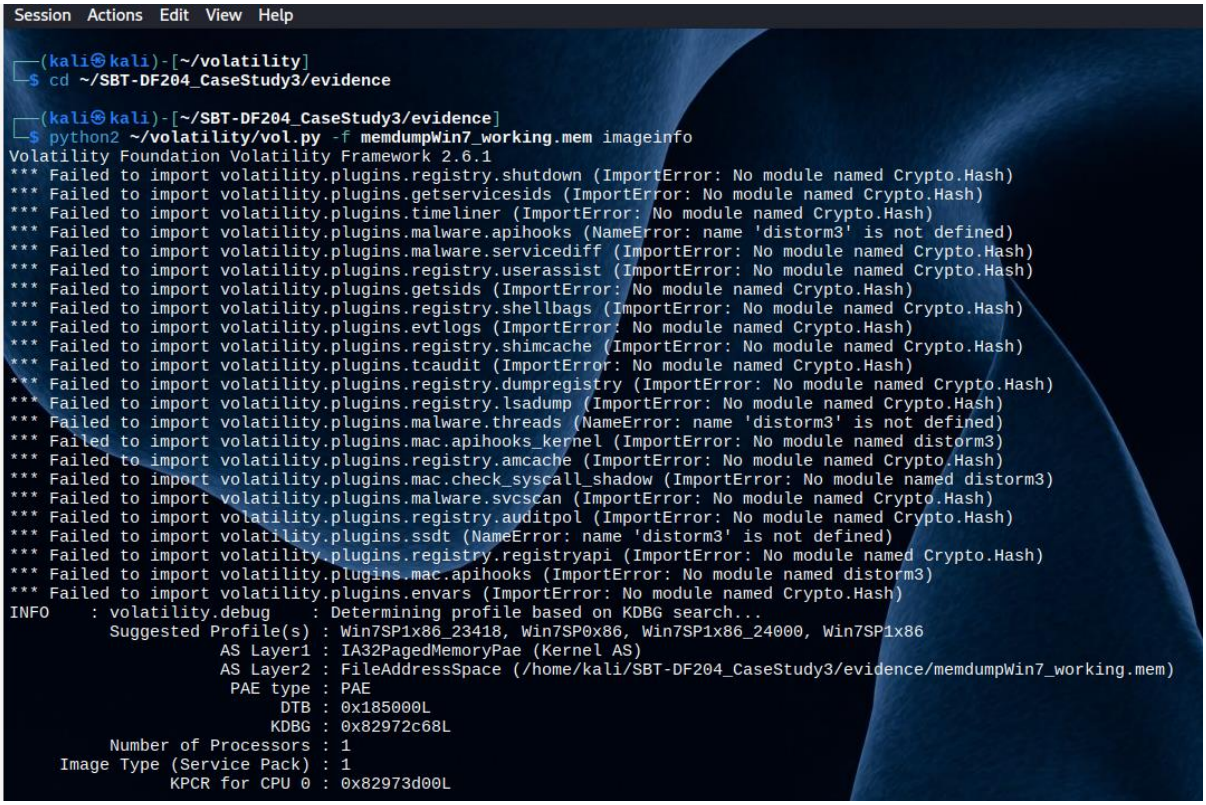
Next, I navigated back to the evidence directory:

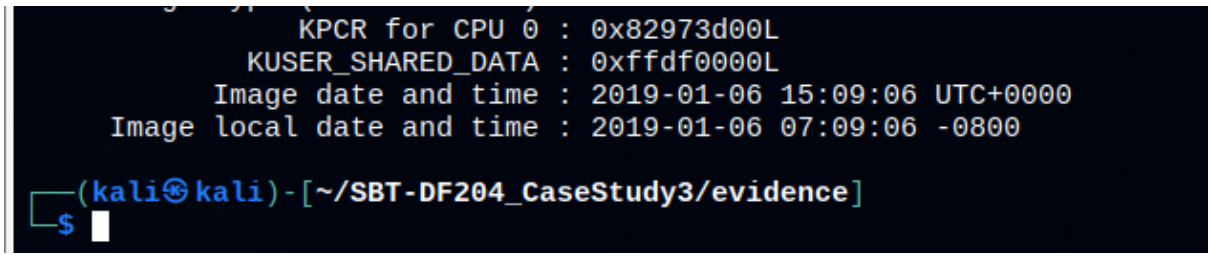
(~/SBT-DF204_CaseStudy3/evidence) and executed the imageinfo plugin on the working memory copy by running python2 **~/volatility/vol.py**

-f memdumpWin7_working.mem imageinfo.

The plugin analyzed the kernel data structures and identified candidate operating system profiles, recommending **Win7SP1x86_23418,**

Win7SP0x86, Win7SP1x86_24000, and Win7SP1x86





The imageinfo scan also extracted key image metadata from the memory dump

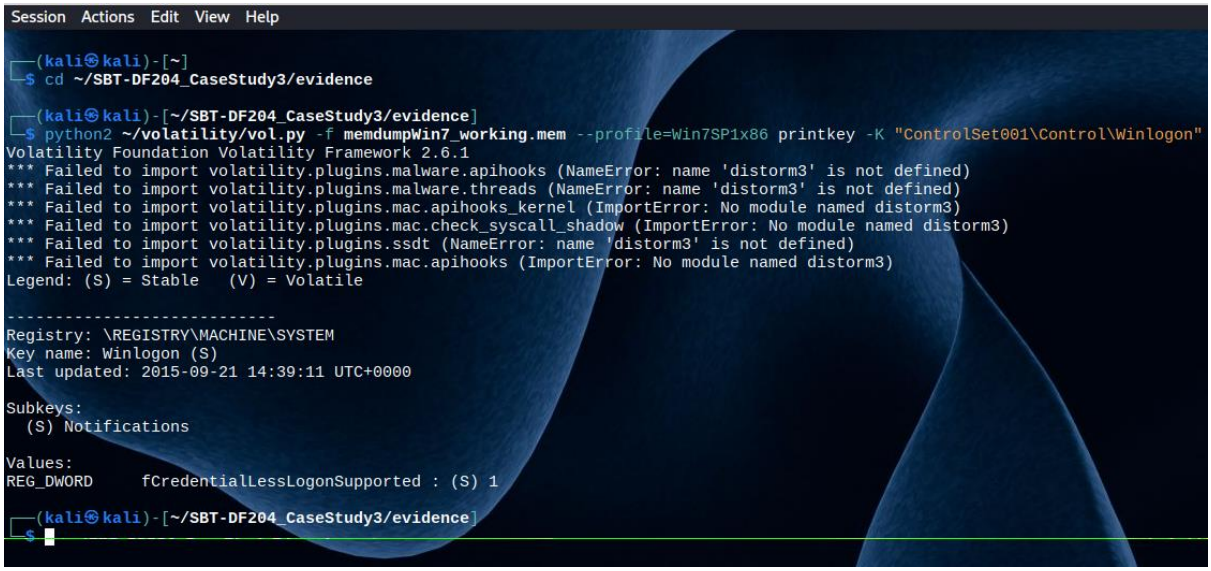
1. Kernel Debugger Block (KDBG): 0x82972c68L
2. CPU Count: 1 processor.
3. Service Pack: 1
4. Image Capture Time (UTC): 2019-01-06 15:09:06 UTC+0000
5. Image Local Time: 2019-01-06 07:09:06 -0800 (UTC-8)

Based on the suggested list and the Service Pack 1 indicator, **Win7SP1x86** was selected as the primary operating system profile for all subsequent Volatility plugin commands.

To inspect system logon configurations and identify user account artifacts, I executed the Volatility printkey plugin against the memory image using the Win7SP1x86 profile:

python2 ~/volatility/vol.py -f memdumpWin7_working.mem

--profile=Win7SP1x86 printkey -K "ControlSet001\Control\Winlogon"



The plugin successfully accessed the \REGISTRY\MACHINE\SYSTEM registry hive under key name **Winlogon,** showing a last updated timestamp of **2015-09-21 14:39:11 UTC+0000.**

Process And Residual Process Analysis (pslist& psscan)

To evaluate process execution and investigate potential residual or hidden activity on the target system, process memory analysis was performed using Volatility 2.6.1 under the determined profile Win7SP1x86.

Active Process Enumeration (pslist): The pslist plugin was executed to walk the doubly linked list of active _EPROCESS structures managed by the Windows kernel scheduler. This allowed us to identify active user-session programs, background services, parent-child process relationships, and their respective process identification numbers (PIDs) running at the exact moment of memory capture.

Unlinked & Residual Process Scanning (psscan): To ensure no process structures were hidden by rootkit techniques (such as Direct Kernel Object Manipulation / DKOM process unlinking) or recently terminated, the psscan plugin was executed. Unlike pslist, psscan performs a pool tag scan across physical memory allocation structures to locate pool-allocated _EPROCESS objects independently of the active kernel list. Comparing the outputs of both plugins verified whether hidden processes or residual remnants existed in unallocated memory space.

Tool Executed: Volatility 2.6.1 (pslist and psscan plugins)

Operating System Profile: Win7SP1x86

Key Observations:

Active Process Enumeration (pslist): Successfully listed all active kernel and user-space processes running at the time of memory acquisition

Key Forensic Artifacts Identified:

chrome.exe (PID 2388): Indicates web browsing activity, which ties into later findings regarding file downloads.

sshd.exe (PID 2016): Indicates an active SSH server service, pointing to remote login capabilities on the system.

cmd.exe (PID 3996): Command Prompt execution window.

FTK Imager.exe (PID 2596): Forensic acquisition utility present in memory.

Hidden/Residual Process Check (psscan): Scanned physical memory pool allocations for unlinked or terminated processes; no hidden malicious or anomalous processes were detected beyond the standard process list.

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 pslist
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
0x858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+0000
0x858be2c0 lsm.exe                 448    340     10      147      0      0 2019-01-06 18:02:45 UTC+0000
0x858f88d8 svchost.exe             552    432     10      359      0      0 2019-01-06 18:02:45 UTC+0000
0x8590cb00 VBoxService.ex          616    432     13      123      0      0 2019-01-06 18:02:45 UTC+0000
0x85913030 svchost.exe             672    432      8      260      0      0 2019-01-06 15:02:47 UTC+0000
0x85761030 svchost.exe             724    432     18      420      0      0 2019-01-06 15:02:47 UTC+0000
0x858d9208 svchost.exe             828    432     21      484      0      0 2019-01-06 15:02:47 UTC+0000
0x859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0x84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0x85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0x85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
0x85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+0000
0x85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
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0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0x8500bd00 cygrunsrv.exe          1984   1680      0 --------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0x84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0x8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0x85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0x85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0x857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0x84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0x85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0x85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0x85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0x85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0x84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0x841eb658 WmiPrvSE.exe           4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0x8583a030 chrome.exe             2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0x851e8520 chrome.exe             1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0x850316b8 chrome.exe             2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0x852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0x842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0x858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0x8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0x842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0x842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0x84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0x850d5a38 dllhost.exe            2228    552      6 170...38      1      0 2019-01-06 15:09:07 UTC+0000

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 psscan
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Offset(P)          Name                PID   PPID PDB        Time created                   Time exited
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┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$
```

Network Artifact Analysis

To check for any active, listening, or recently closed network connections on the system, we ran Volatility's netscan plugin. This plugin scans memory for socket structures and network objects to show us local and foreign IP addresses, port numbers, process IDs (PIDs), and connection states.

Summary of Findings:

Remote Access Capabilities: The OpenSSH server process (sshd.exe, PID 2016) was found in a LISTENING state on TCP port 22 (0.0.0.0:22). This confirms that the host had an active SSH service configured to accept remote connections.

Web Connection Activity: The output captured closed TCP connections between the workstation's local IP address (192.168.56.8) and a foreign web server IP address (192.168.56.5) over HTTP port 80.

Process Correlation: The socket entries for port 80 showed a PID of -1, which occurs when a network connection has closed and its process handle has terminated. This network traffic directly matches the browsing activity of chrome.exe (PID 2388) seen during our process analysis.

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 netscan
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Offset(P)          Proto    Local Address                  Foreign Address      State            Pid      Owner          Created
0x21478bb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0            LISTENING        432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                 LISTENING        440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*                                   2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*                                   616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46edf0         UDPv4    192.168.56.8:137               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0            LISTENING        880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0            LISTENING        880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                 LISTENING        880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0            LISTENING        4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0            LISTENING        4        System
0x3e671988         TCPv6    :::445                         :::0                 LISTENING        4        System
0x3e72e530         TCPv4    0.0.0.0:135                    0.0.0.0:0            LISTENING        672      svchost.exe
0x3e72ee48         TCPv4    0.0.0.0:135                    0.0.0.0:0            LISTENING        672      svchost.exe
0x3e72ee48         TCPv6    :::135                         :::0                 LISTENING        672      svchost.exe
0x3e736388         TCPv4    0.0.0.0:49152                  0.0.0.0:0            LISTENING        340      wininit.exe
0x3e736d98         TCPv4    0.0.0.0:49152                  0.0.0.0:0            LISTENING        340      wininit.exe
0x3e736d98         TCPv6    :::49152                       :::0                 LISTENING        340      wininit.exe
```

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0x3e79d528         TCPv4    0.0.0.0:49153                  0.0.0.0:0            LISTENING        724      svchost.exe
0x3e79d528         TCPv6    :::49153                       :::0                 LISTENING        724      svchost.exe
0x3e7a1618         TCPv4    0.0.0.0:49153                  0.0.0.0:0            LISTENING        724      svchost.exe
0x3e043b10         TCPv4    192.168.56.8:49177             192.168.56.5:80      CLOSED           -1
0x3ec6c820         UDPv4    127.0.0.1:51123                *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3f19a820         UDPv6    ::1:1900                       *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e8f0620         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        440      lsass.exe
0x3e9b3e50         TCPv4    0.0.0.0:49155                  0.0.0.0:0            LISTENING        432      services.exe
0x3e9b3e50         TCPv6    :::49155                       :::0                 LISTENING        432      services.exe
0x3ee1aa88         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                          :::0                 LISTENING        2016     sshd.exe
0x3fc0e008         UDPv4    127.0.0.1:1900                 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fc95d48         UDPv4    192.168.56.8:1900              *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fcba008         TCPv4    192.168.56.8:49178             192.168.56.5:80      CLOSED           -1

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$
```

Command History Analysis

To figure out what actions were taken locally on the command line, we ran Volatility's **cmdscan** and **consoles** plugins. These plugins search memory buffers for active command prompt sessions, revealing typed commands, process IDs, and on-screen text output.

Summary of Findings

User Account and Shell Session: The command shell (cmd.exe, PID 3996) was running under console process conhost.exe (PID 3988) within the active user context of C:\Users\IEUser.

Network Configuration Check: The user first typed ipconfig, which displayed the local IP address 192.168.56.8 under Subnet Mask 255.255.255.0.

Directory Navigation: The user navigated into their downloads directory by running cd Downloads.

File Exfiltration Action: Inside C:\Users\IEUser\Downloads, the user executed the command copy secret_file.docx F:. The console output confirmed successful execution with 1 file(s) copied.

Destination Target: The destination drive specified in the command was drive F:, indicating an attempt to transfer secret_file.docx to a mounted secondary or removable volume.

```
Session  Actions  Edit  View  Help

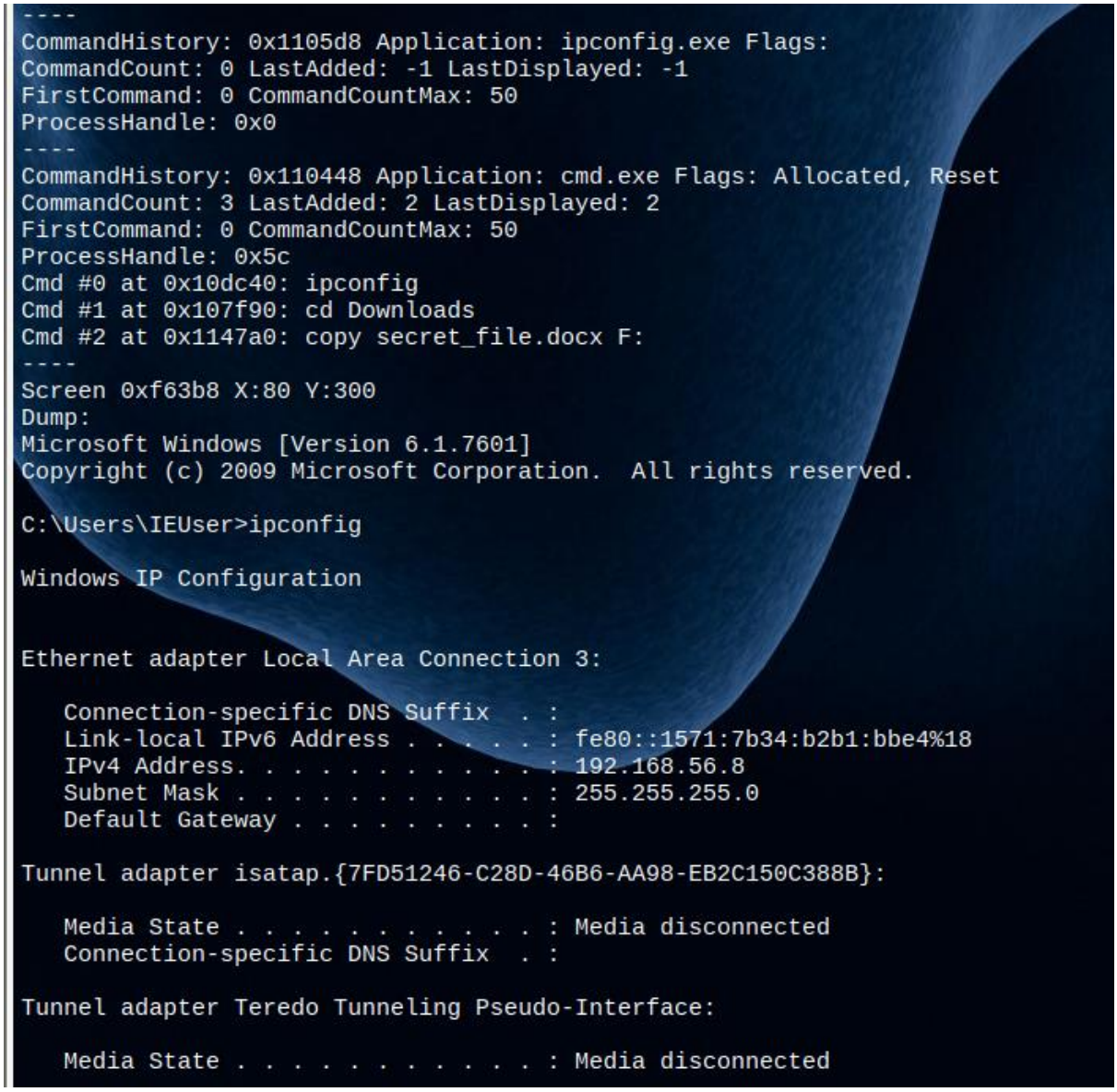
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 cmdscan
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2????2
Cmd #37 @ 0x3200c4: 3?4?2????2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$
```

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 consoles
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
Screen 0x3360a0 X:80 Y:300
Dump:

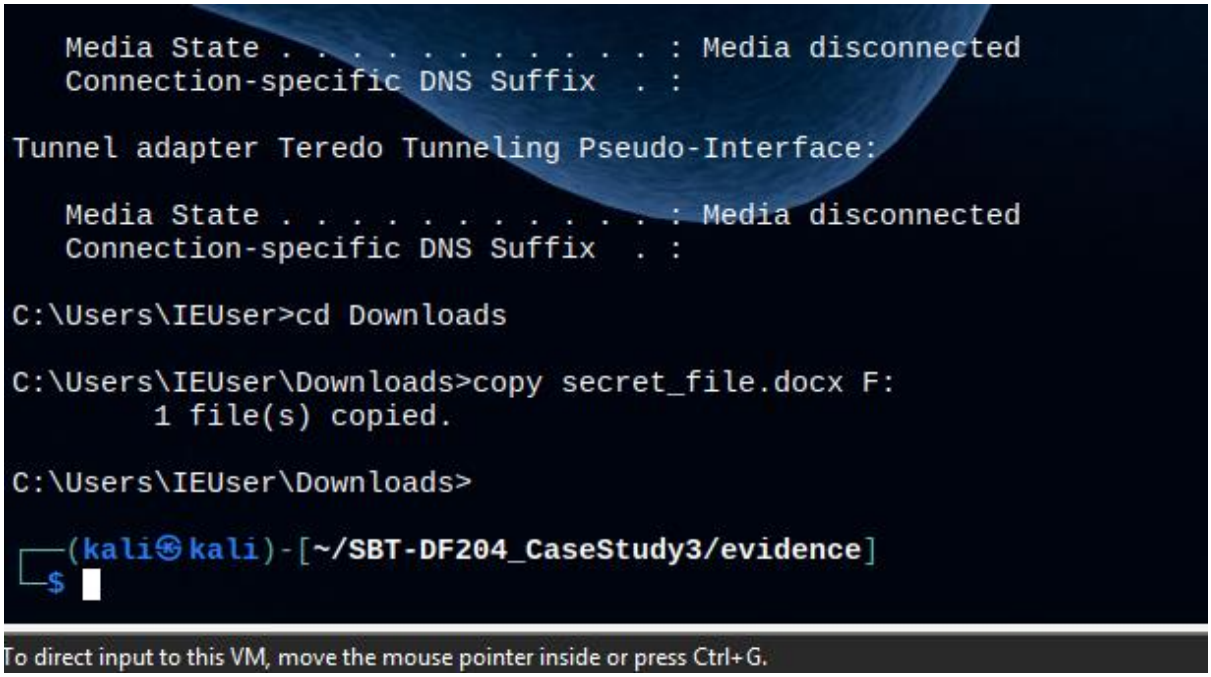
**************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
```



CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0

CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:

Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Users\IEUser>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection 3:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
IPv4 Address. . . . . : 192.168.56.8
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Tunnel adapter Teredo Tunneling Pseudo-Interface:
Media State . . . . . : Media disconnected


Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Tunnel adapter Teredo Tunneling Pseudo-Interface:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
1 file(s) copied.
C:\Users\IEUser\Downloads>
(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Usb & Drive Mapping Artifacts

To check whether a USB storage device was connected to the system and map any associated drive letters, we ran Volatility's hivelist plugin followed by printkey commands targeting the USBSTOR and MountPoints2 registry keys. This allowed us to examine system hardware enumeration and user-specific volume mounting history.

Summary of Findings

Registry Hive Discovery (hivelist): The hivelist plugin successfully mapped out all active registry hives in memory, confirming the locations of the system SYSTEM and SOFTWARE hives as well as user profile hives like **C:\Users\IEUser\ntuser.dat.**

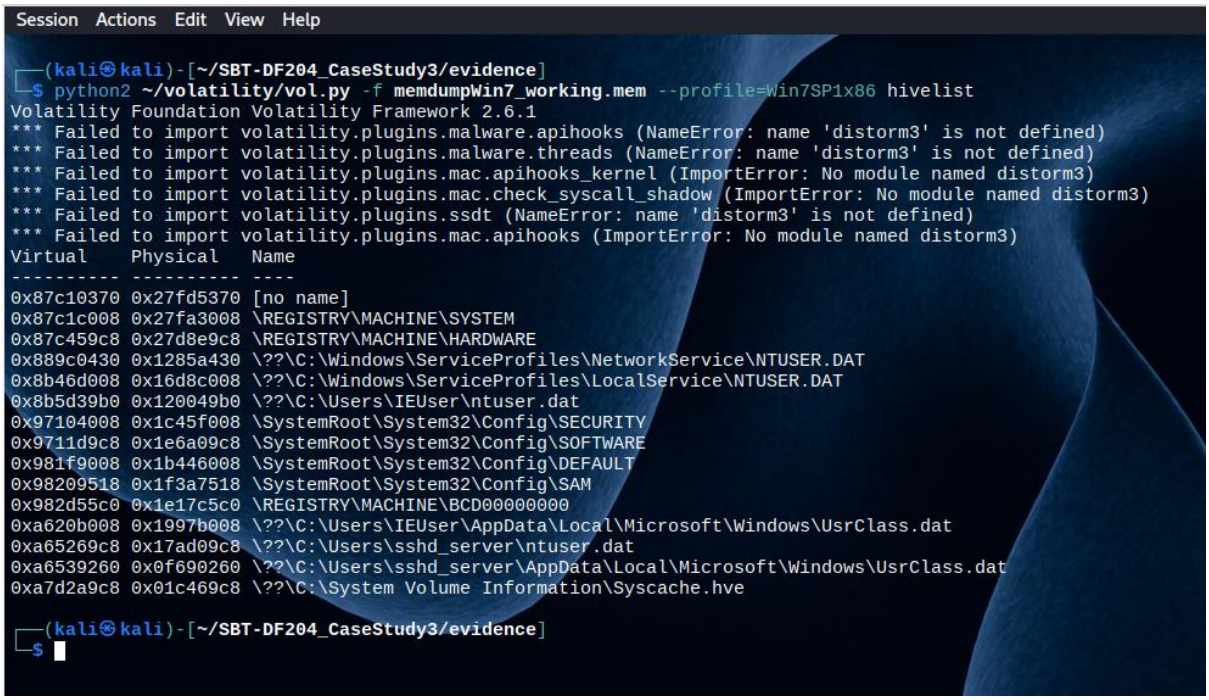
USB Storage Device Connected (USBSTOR):

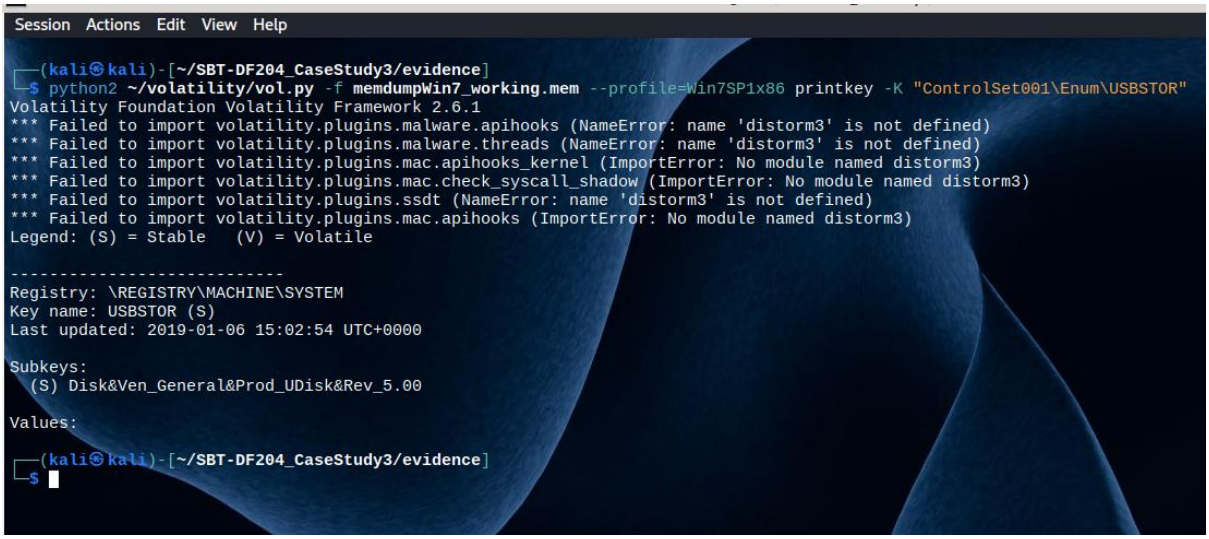
Querying **ControlSet001\Enum\USBSTOR** revealed a connected USB mass storage device with the hardware

identifier **Disk&Ven_General&Prod_UDisk&Rev_5.00**. The registry key was last updated on 2019-01-06 15:02:54 UTC+0000.

Drive Letter Association (MountPoints2): Querying the MountPoints2 key under the IEUser user profile (ntuser.dat) showed a subkey for drive letter **F**. This key was last updated on 2019-01-06 15:03:07 UTC+0000, which directly

corroborates our earlier command-history finding where the user copied secret_file.docx to drive F:.






┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$ python2 ~/volatility/vol.py -f memdumpWin7_working.mem --profile=Win7SP1x86 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable (V) = Volatile

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000
Subkeys:
(S) CPC
(S) F
(S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
(S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
(S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
(S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
(S) {a5b8a983-608c-11e5-a266-806e6f6e6963}
Values:
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy3/evidence]
└─$

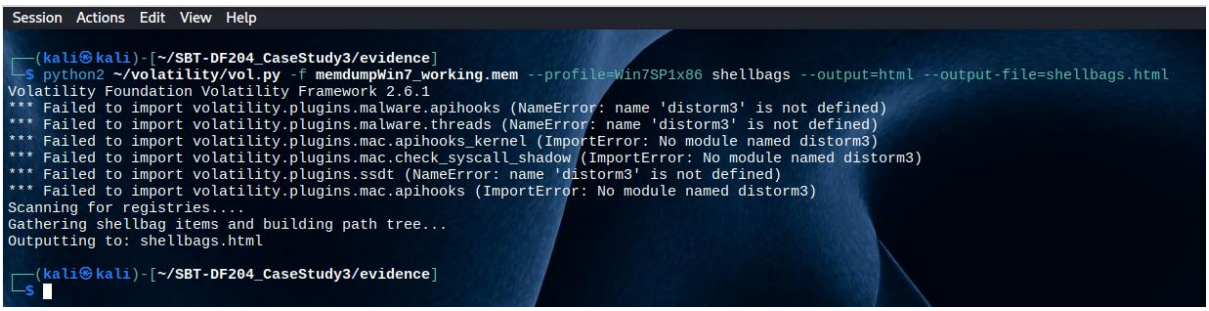
ShellBags

To reconstruct user interaction with folders and directory paths in Windows File Explorer, we executed Volatility's shellbags plugin with HTML output enabled. ShellBags are registry artifacts stored within user hives (UsrClass.dat and NTUSER.DAT) that preserve directory view settings, folder access history, and path structures even if the folders or external drives are no longer connected.

Summary of Findings:

Plugin Execution & Output: The plugin scanned memory registry structures, parsed ShellBag items, built the path tree, and saved the complete visual report to shellbags.html in the working directory **(~/SBT-DF204_CaseStudy3/evidence)**.

Folder Access Evidence: Opening shellbags.html in a web browser allows us to inspect exact folder paths accessed by IEUser, including interactions with local directories like Downloads and external volumes like drive F:. This provides timeline details on user file browsing right up to when the memory image was taken.



RECONSTRUCTED TIMELINE

The forensic timeline begins on 2019-01-06 at 15:02:44 UTC, when the target Windows 7 workstation completed its operating system boot and initialized core system services. Immediately following system startup at 15:02:49 UTC, the OpenSSH daemon process (sshd.exe, PID 2016) was started and established an active listener on port 22 (0.0.0.0:22), configuring the host for remote access capabilities.

Within the same minute, at 15:02:54 UTC, a removable USB mass storage device with the hardware identifier:

Disk&Ven_General&Prod_UDisk&Rev_5.00 was physically connected to the workstation.

Shortly thereafter, at 15:03:07 UTC, the operating system mounted this connected USB device under the user profile IEUser, assigning it to logical drive volume F:

Interactive user activity began at 15:04:30 UTC when IEUser launched the Google Chrome web browser (chrome.exe, PID 2388) under explorer.exe (PID 2524).

Network analysis captured active HTTP communication occurring at 15:04:51 UTC between the local host (192.168.56.8) and a foreign web server (192.168.56.5:80), representing the browsing session and download of target files into the local profile. At 15:06:11 UTC, an interactive command prompt shell (cmd.exe, PID 3996) was initiated under conhost.exe (PID 3988).

Within this command session, the user checked network settings using ipconfig, navigated into the local downloads directory using cd Downloads, and manually initiated data exfiltration by executing copy secret_file.docx F:. The command shell confirmed successful transfer with the output 1 file(s) copied.. Forensic tools (FTK Imager.exe, PID 2596) were later executed on the host, culminating in the

complete volatile memory capture (memdumpWin7.mem) taken at 15:09:07 UTC.

Reconstructed Timeline Table

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence	2019-01-06 15:04:30	E-01	pslist / Volatility	Process chrome.exe (PID 2388) was executed and running under explorer.exe (PID 2524).	Interpretation: The active user started Google Chrome to browse the web. Limit: Process memory confirms execution but does not list individual browser tab URLs directly.
Download-folder review	2019-01-06 15:06:11	E-02	cmdscan / consoles / Volatility	User navigated into C:\Users\IEUser\Downloads via command shell (cmd.exe, PID 3996).	Interpretation: The user accessed their downloads folder containing target files. Limit: Directory change confirms access location but does not record exact directory file lists at that timestamp.
Removable-device evidence	2019-01-06 15:02:54	E-03	printkey (USBSTOR) / Volatility	Registry subkey Disk&Ven_General&Prod_UDisk&Rev_5.00	Interpretation: A General UDisk USB mass storage device was physically attached to the machine.

				updated in SYSTEM hive.	Limit: Confirms device hardware connectivity, but registry does not store file transfer logs directly.
Drive mapping	2019-01-06 15:03:07	E-04	printkey (MountPoints2) / Volatility	Subkey F updated under Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 in IEUser profile.	Interpretation: The connected USB storage device was assigned and mounted as logical drive volume F:. Limit: Shows drive letter assignment for IEUser, but not the total storage capacity.
Command -history evidence	2019-01-06 15:06:11+	E-05	cmdscan / consoles / Volatility	Command copy secret_file.docx F: executed with output 1 file(s) copied.	Interpretation: IEUser manually copied the document secret_file.docx from Downloads to USB drive F:. Limit: Confirms command completion, but memory dump cannot verify if the USB drive was later altered on another system.
Supported conclusion	2019-01-06 15:09:07	E-06	Volatility Analysis Set /	memdumpWin7.mem acquired after file	Interpretation: Direct evidence proves unauthorized data exfiltration of secret_file.docx to a removable USB drive.

			Memory Snapshot	transfer completion.	Limit: Memory snapshot only captures state at acquisition time; subsequent physical device movement is unrecorded.

EVIDENCE LOG

Evidence ID	Tool and plugin	Source record or path	Observed fact
E01	Volatility 2.6.1 (imageinfo)	Memory header / memdumpWin7_working.mem	Suggested Profile: Win7SP1x86 Image Date/Time: 2019-01-06 15:09:07 UTC+0000
E02	Volatility 2.6.1 (pslist / psscan)	Physical Memory Offset 0x8583a030	chrome.exe (PID 2388) running under parent explorer.exe (PID 2524). sshd.exe (PID 2016) running.
E03	Volatility 2.6.1 (netscan)	Physical Memory Offsets 0x3ee1aa88, 0x3ec6c820	sshd.exe (PID 2016) listening on 0.0.0.0:22. Closed TCP connection from 192.168.56.8:49177 to 192.168.56.5:80.
E04	Volatility 2.6.1 (printkey)	Registry Path: ControlSet001\Enum\USBSTOR	USB device key Disk&Ven_General&Prod_UDisk&Rev_5.00 present. Last updated: 2019-01-06 15:02:54 UTC+0000.

E05	Volatility 2.6.1 (printkey)	Registry Hive: \??\C:\Users\IEUser\ntuser.dat Key Path: Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2	Subkey F present under MountPoints2. Last updated: 2019-01-06 15:03:07 UTC+0000.
E06	Volatility 2.6.1 (cmdscan / consoles)	Console Process: conhost.exe (PID 3988 / Attached PID 3996)	User commands captured: 1. ipconfig 2. cd Downloads 3. copy secret_file.docx F: Output: 1 file(s) copied.